

KnowSelf

Agentic Knowledgeable Self-Awareness for LLM-Based Agents

Teaching LLM Agents When to Think, Reflect, or Seek Knowledge

Enabling agents to autonomously regulate knowledge utilization through situational self-awareness.

Source: arxiv.org/abs/2504.03553

Current Agent Training Uses 'Flood Irrigation'

Indiscriminate Knowledge Injection Hurts Efficiency

Current agent learning resembles unconscious pattern-fitting rather than situational awareness. They blindly inject knowledge regardless of whether the agent needs it at a given step.



Trajectory Imitation

e.g., ReAct

Direct imitation without assessing capability.



Trial-and-Error

e.g., Reflexion

Inefficiently tries and fails before adapting.



Knowledge-Augmented

e.g., KnowAgent, WKM

Injects domain knowledge indiscriminately at every step.

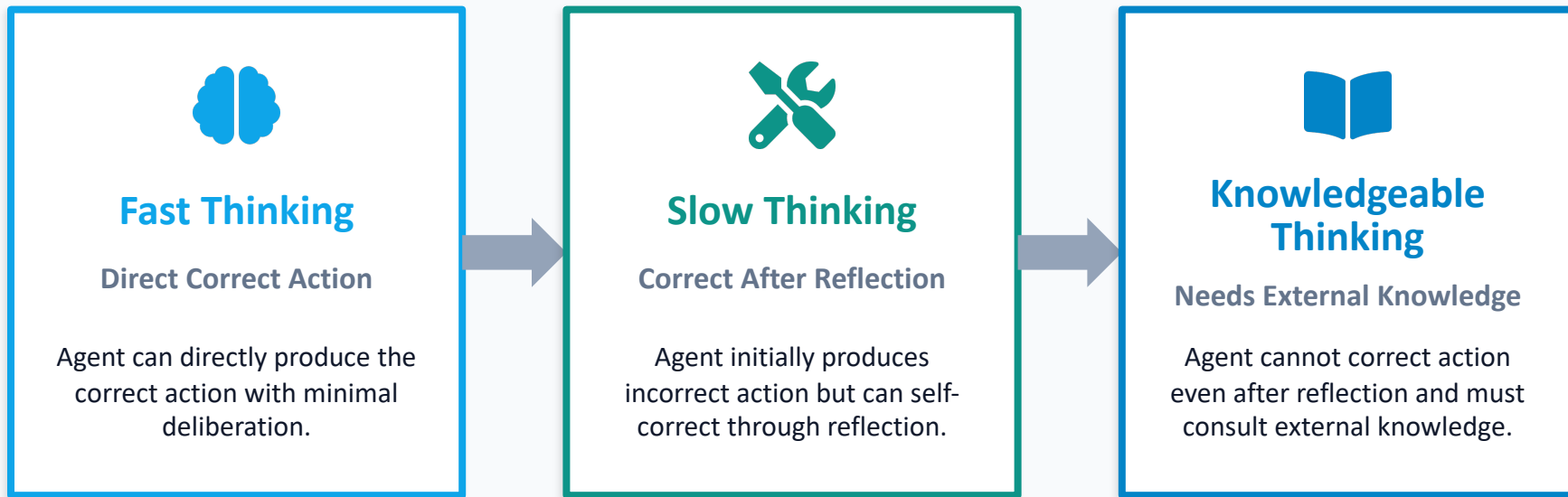


The Solution: Human-like Situational Self-Awareness

Metacognitive ability to assess whether to handle directly, reflect, or seek external help.

Three Situational Modes

Fast Thinking, Slow Thinking, and Knowledgeable Thinking



KnowSelf Framework

Data Construction → Two-Stage Training → Self-Aware Inference

Step 1: Self-Awareness Data Construction

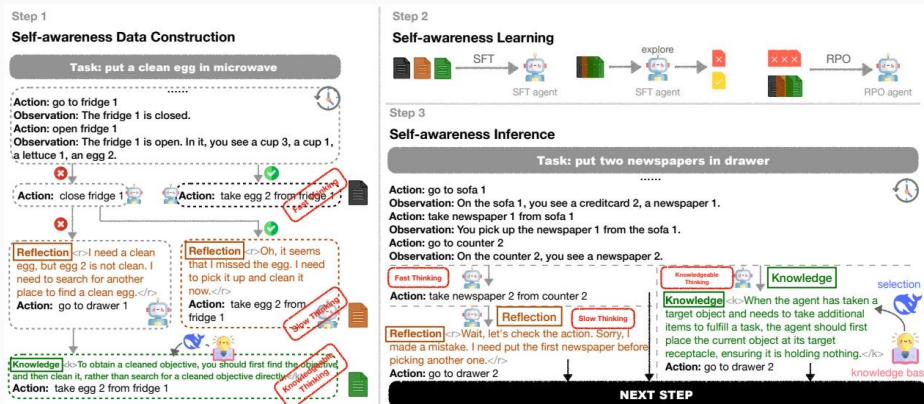
Heuristic criterion classifies situation to mark self-explored trajectories with special tokens.

Step 2: Two-Stage Training

First SFT teaches initial patterns, followed by Reward-based Policy Optimization (RPO) to refine token generation.

Step 3: Self-Aware Inference

Deployment agent uses generated tokens to dynamically trigger direct action, reflection, or knowledge retrieval.



KnowSelf Correctly Heats a Mug & Places a Saltshaker

Where Baselines Fail

(a) Task: put a hot mug in cabinet

Action: take mug 1 from countertop 1
Observation: You pick up the mug 1 from the countertop 1.
Action: go to microwave 1
Observation: The microwave 1 is closed.
Action: open microwave 1
Observation: You open the microwave 1. In it, you see an egg 1.
[Knowledge]<k>If the agent is in a situation where it needs to modify the temperature of an object using a receptacle (e.g., a fridge for cooling), it should directly perform the specific action like 'cool' or 'heat' instead of placing the object on it.</k>
Action: take egg 1 from microwave 1
Observation: Nothing happens.

OpenAI-O1

Action: take mug 1 from countertop 1
Observation: You pick up the mug 1 from the countertop 1.
Action: go to microwave 1
Observation: The microwave 1 is closed.
Action: open microwave 1
Observation: You open the microwave 1. In it, you see an egg 1.
Action: put mug 1 in/on microwave 1
[Reflection]<r>Wait, let's check the action. Let's think step by step. With the microwave open, I can heat the mug using the microwave.</r>
Action: heat mug 1 with microwave 1

KnowSelf
(Llama-8B)

(b) Task: put a saltshaker in drawer

Action: take saltshaker 1 from countertop 3
Observation: You pick up the saltshaker 1 from the countertop 3.
Action: go to drawer 1
Observation: The drawer 1 is open. In it, you see a fork 1.
Action: put saltshaker 1 in/on drawer 1
[Reflection]<r>Wait, the drawer is open, so the saltshaker might fall inside. I need to close the drawer first.</r>
Action: close drawer 1
Observation: You close the drawer 1.
Action: put saltshaker 1 in/on drawer 1
Observation: Nothing happens.

DeepSeek-R1

Action: take saltshaker 1 from countertop 3
Observation: You pick up the saltshaker 1 from the countertop 3.
Action: go to drawer 1
Observation: The drawer 1 is closed.
Action: open drawer 1
Observation: You open the drawer 1. In it, you see a fork 1.
[Knowledge]<k>When the agent encounters a situation where receptacles contain other items but only needs to place the target object, the agent should proceed to place the object without removing unrelated items.</k>
Action: put saltshaker 1 in/on drawer 1

KnowSelf
(Llama-8B)

KnowSelf **Example A: 'put a hot mug in cabinet'** KnowSelf uses reflection to heat the mug while OpenAI-O1 gets confused

KnowSelf Achieves 84.33% on ALFWorld

with Llama-8B

Using Only 15% Knowledge

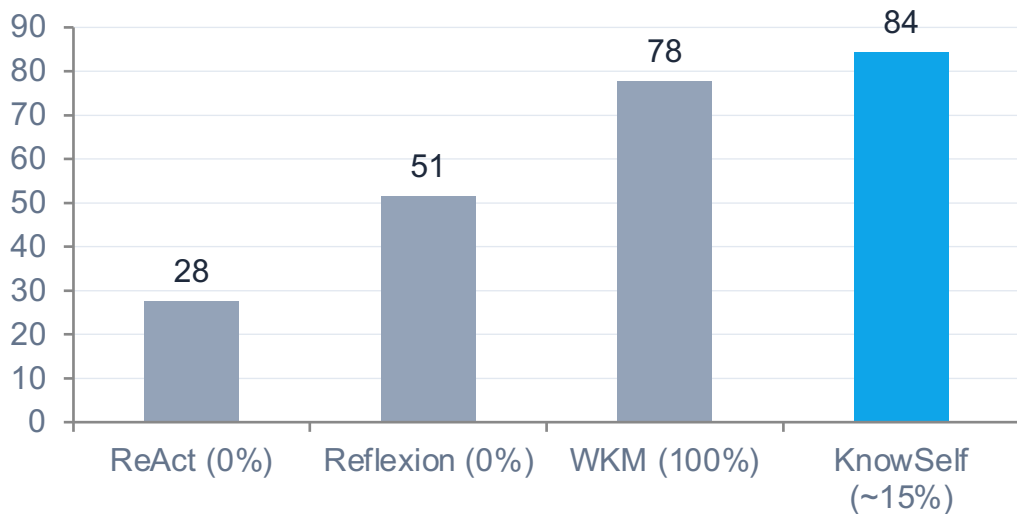
Backbone	Method	Know%	All	Put	Clean	Heat	Cool	Examine	PutTwo
GPT-4o	ReAct	0%	76.12	83.33	74.19	69.57	85.71	77.78	64.71
GPT-4o	Reflexion	0%	85.07	100.00	87.10	73.91	90.48	83.33	70.59
GPT-4o	ExpeL	100%	79.85	95.83	83.87	69.57	80.95	88.89	52.94
Llama-8B	ReAct	0%	27.61	33.33	3.23	0.00	57.14	66.67	23.53
Llama-8B	Reflexion	0%	51.49	62.96	22.73	5.88	64.29	86.36	50.00
Llama-8B	ETO	0%	78.36	91.67	70.59	82.61	61.90	88.89	64.71
Llama-8B	KnowAgent	100%	75.37	87.50	93.55	65.22	66.67	61.11	64.71
Llama-8B	WKM	100%	77.61	95.83	87.10	86.96	61.90	66.67	52.94
Llama-8B	KnowSelf	15.01%	84.33	91.67	87.10	91.30	85.71	77.78	64.71
Gemma-2B	ETO	0%	71.64	91.67	83.87	78.26	52.38	77.78	29.41
Gemma-2B	WKM	100%	76.12	91.67	87.10	78.26	71.43	61.11	52.94

KnowSelf uses far less knowledge (15% and 26%) yet outperforms methods using 100% knowledge, maintaining strong scaling behavior down to Gemma-2B models.

Minimal Knowledge, Maximum Impact

Efficiency Advantage Over Full-Knowledge Methods

Overall Accuracy (%) - Llama-8B



15% vs 100%

Knowledge Utilization

KnowSelf uses a fraction of the queries, **reducing API costs and inference time** while achieving state-of-the-art scores.

Key Takeaways

Self-Awareness Is the Missing Piece in Agent Planning

- **Wasteful paradigms:** Indiscriminate knowledge injection is suboptimal.
- **Learnable self-awareness:** Agents learn situational metacognition via special tokens with lightweight heuristic labeling.
- **State-of-the-art performance:** KnowSelf (Llama-8B) outperforms strong baselines like GPT-4o ReAct (84.33% vs 76.12%).
- **Unmatched efficiency:** Requires only 15-26% knowledge usage compared to 100% for competing methods.
- **Generalization across scales:** Paradigm works even for smaller models (Gemma-2B reaches 79.85%, matching GPT-4o ExpEL).